

SMARTROAD BACKEND GUIDE

Purpose

The backend stores road issue reports centrally and serves the site.

Run

1. Install Python 3.10 or newer.
2. In the project folder: `python backend/server.py`
3. Open `http://127.0.0.1:8000`

Data

Reports persist in `backend/data/reports.json`.

The frontend loads from the API and saves changes to it automatically.

If the API is unavailable, browser `localStorage` remains a fallback.

Endpoints

GET `/api/health` - health check

GET `/api/reports` - list reports; supports `problemType`, `severity`, `status`

POST `/api/reports` - create a report

PUT `/api/reports` - replace collection (used by current UI)

GET, PATCH, DELETE `/api/reports/{id}` - manage one report

DELETE `/api/reports` - clear all reports

GET `/api/stats` - dashboard totals

Safety

The API validates issue values, required text, coordinate types and images.

Images are capped at 2 MB. For production use a database, object storage, authentication, restricted CORS, and HTTPS.